

All of these reasons point to viruses as the future of hostile algorithms, however, the most significant reason for this determination is the effectiveness of the virus as a form of attack. Past experiments by Doctor Fred Cohen [1984] used a normal user account on a Unix system, without privileged access, and gained total security penetration in 30 minutes. Doctor Cohen repeated these results on many versions of Unix, including AT&T Secure Unix and over 20 commercial implementations of Unix. The results have been confirmed by independent researchers worldwide. Separate experiments by Tom Duff [1989] demonstrated the tenacity of Unix viruses even in the face of disinfectors. The virus used in Mr. Duff's experiment was a simple virus written in script. The virus was believed to have been reintroduced by the operating system from the automated backup and restore system. Re-infection took place after the system had been virus free for one year.

Heterogeneous Virus Attacks

Non-Unix PCs attached to a heterogeneous network that were infected with computer viruses originating from Unix servers have been observed. The Unix systems were not the original point of entry for the viruses. They were dormant while on the Unix systems but became harmful when they migrated to their target systems. The Unix systems acted as unaffected carriers of computer viruses for other platforms. For the sake of simplicity, I have named this effect after an historical medical problem of similar nature, 'Typhoid Mary Syndrome'. Networks and specifically Unix servers that provide network file systems are very susceptible to this problem.

This problem was first observed while investigating an infection of personal computers attached to a network with a large population of Unix servers and workstations. The virus was manually attacked on the personal computers using virus scanners. During the infection period all of the personal computers were disconnected from the network and idle. Once all the computers were disinfected, all removable media was tested and the infection was unobserved for a period of time, the